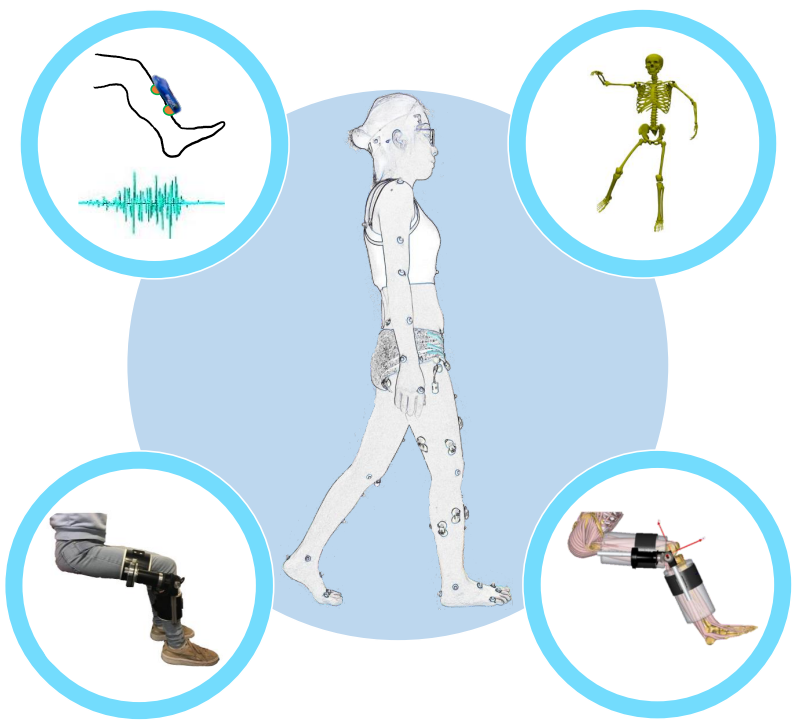
KTH ROYAL INSTITUTE
OF TECHNOLOGY

Doctoral Thesis in Engineering Mechanics

Biomechanical Parameter Estimation for Wearable Exoskeleton System Design

LONGBIN ZHANG



Stockholm, Sweden 2021

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Academic Dissertation which, with due permission of the KTH Royal Institute of Technology, is submitted for public defence for the Degree of Doctor of Philosophy onFriday, the 22nd October 2021, at 9:00 a.m. in F3, Lindstedsvägen 26, Stockholm.

Doctoral Thesis in Engineering Mechanics
KTH Royal Institute of Technology
Stockholm, Sweden 2021

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ISBN 978-91-8040-021-3
TRITA-SCI-FOU 2021:38

Printed by: Universitetsservice US-AB, Sweden 2021

To my parents

Biomechanical parameter estimation for wearable exoskeleton system design

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Abstract

Exoskeletons are increasingly used in rehabilitation and daily life in persons with motor disorders after neurological injuries. The overall objective of this thesis is to study how to robustly and accurately predict joint torque using inputs from sensors that would technically be feasible to equip on an assistive exoskeleton, and to develop a framework that could be used to evaluate the user-exoskeleton interface. This compilation thesis is based on five papers that focus on different aspects of exoskeleton controller design and physical design, to contribute to the design of wearable exoskeleton systems for rehabilitation and movement assistance.

In the first study, we estimated ankle joint torques using an electromyography (EMG)-driven neuromusculoskeletal (NMS) model and a standard artificial neural network (ANN) model in seven movement tasks. EMG signals and ankle joint angles were used as input in each method, the models were trained with experimental data from 3D motion analysis, and ankle joint torques were predicted. Our results suggested that the standard-ANN model could predict ankle torques more accurately when trained on a large and varied set of trials but less accurately in unseen movements whereas the NMS model was overall more useful and has the advantage of detailing underlying physical principles.

The second study was directly inspired by the first study. Our aim was to estimate ankle joint torques using a hybrid form of the two methods in the first study, specifically an NMS solver-informed ANN, by augmenting a standard-ANN model with additional features from the underlying NMS solver. The torque prediction performance of hybrid-ANN models was investigated in two scenarios: intra-subject and inter-subject tests. We found that the NMS solver informed-ANN model had a better joint torque prediction accuracy than the methods independently (i.e. NMS or the standard-ANN model), which indicates that benefit was gained from integrating NMS features into standard ANN models. Furthermore, in the inter-subject tests, we adopted the transfer learning technique to further improve the joint torque performance by taking advantage of the information extracted from previous subjects.

In the third study, we predicted not only ankle joint torques, but also knee joint torques in the sagittal plane and hip joint torques in both sagittal and frontal planes during several different motions using long short-term memory memory (LSTM) neural networks. Several networks were constructed, namely

separate LSTM models for each movement as well as one uniform model for all movements. The LSTM networks were constructed both with and without transfer learning, in which the networks could take advantage of information extracted from previous tasks and/or subjects to predict for an unseen or "new" task or subject. We evaluated the LSTM models' prediction ability across tasks and subjects, and studied whether generalizability was improved when transfer learning was implemented in the LSTM networks. Our results indicate that both one uniform and ten separate LSTM models predicted lower limb joint torques accurately, i.e. with low prediction error, in intra-subject tests. By including transfer learning, the LSTM models' generalizability in predicting moments across tasks and subjects was significantly improved, even if re-trained with a smaller subset of movements from the "new" subject.

In the fourth and fifth studies, we generated a realistic knee exoskeleton model. In the fourth study, we created a virtual human-machine system (HMS) in a musculoskeletal modeling software. We found that this proposed virtual HMS was useful for simulating how different assistive strategies may affect parameters that describe the demands and consequences on the user, including muscular demand, joint contact forces and human-machine interactive forces involved in movements. In the fifth study, we analyzed, through simulation, how different weight distributions of knee exoskeleton components influenced the user's muscle activation during three functional movements. One unilateral knee exoskeleton prototype was fabricated and tested on five healthy subjects. Simulation and experimental muscle activations were compared. We found that muscle activation varied among weight distributions and movements, indicating that no single physical design was optimal for all movements. Exoskeleton physical design should ideally consider the user's activity goals.

This methods developed in this thesis hold great promise in applications such as the design of assistance-as-needed exoskeleton control. The developed virtual HMS makes it possible to study how a knee exoskeleton's different assistive strategies and physical designs can influence its user's muscle effort and interaction forces, which can both simplify prototype iterations and evaluate some parameters that are otherwise difficult to measure experimentally.

Keywords: Hill-type muscle model, generalization, transfer learning, computationally modeling, human-machine system

Biomekanisk parameteruppskattning för bärbar exoskeletonsystemdesign

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Sammanfattning

Exoskelett används alltmer vid rehabilitering och dagligt liv hos personer med motoriska störningar efter neurologiska skador. Det övergripande syftet med denna avhandling är att skapa robusta och noggranna metoder för att förutsäga vridmomentet i nedre extremiteterna med bärbara sensorer samt för att utvärdera i en simulering krafterna i gränssnittet mellan ett exoskelett och en användare. Denna avhandling grundar sig på fem studier var fokus är på olika aspekter av en exoskeletts reglerteknik och fysisk design, i en kontext för tillämpning i framtagning av exoskelett för rehabilitering och rörelseassistans.

I den första studien har vi uppskattat vridmomentet i fotleden i sju olika rörelser med hjälp av en elektromyografi (EMG) driven neuromuskuloskeletal (NMS) modell och en standard artificiell neuralt nätverk (ANN) modell. EMG-signaler och fotledsvinklar användes som input i både modeller. Modellerna tränades med experimentella data från 3D-rörelseanalys för att sedan kunna uppskatta vridmoment med enbart inputdata. Våra resultat tyder på att ANN modellen kunde uppskatta vridmoment noggrannare än NMS modellen om den tränades på mycket data från många typer av rörelser; men NMS modellen uppskattade noggrannare när den tränades med data från liknande rörelser. Dessutom har NMS modellen fördelen av att kunna beskriva underliggande fysiska principer.

I den tredje studien uppskattade vi vridmoment i fotleden och i knäleden i sagittalplan, samt i höftleden i både sagittal- och frontalplan, under flera olika rörelser med hjälp av ett LSTM neuralt nätverk. Flera nätverk konstruerades, nämligen separata LSTM modeller för varje rörelse samt en enhetlig modell för alla rörelser. LSTM nätverken konstruerades både med och utan s.k. “transfer learning”, där ett nätverk kan dra nytta av kunskap från tidigare rörelser och/eller försökspersoner för att bättre uppskatta vridmoment hos en tidigare okänd eller “ ny ” rörelse eller testperson. Vi utvärderade hur noggrant som LSTM modellerna kunde uppskatta vridmoment när de tränades och testades på olika rörelser och försökspersoner. En av studiefrågorna var huruvida modellernas generaliserbarhet förbättrades när transfer learning implementerades.

Våra resultat tyder på att både den enhetliga och de tio individuella LSTM-modellerna kunde noggrant uppskattade vridmoment. Transfer learning ytterligare ökade deras generaliserbarhet för nya rörelser och försökspersoner, även om dataset för träning från den “nya” testpersonen innehöll mindre data från färre rörelser.

I den fjärde och femte studien skapade vi en realistisk knäexoskeletmodell. I den fjärde studien skapade vi ett virtuellt human-machine system (HMS) i biomekanikmodellering mjukvara. Vi demonstrerade att HMS modellen var till stor nytta genom att simulera hur olika strategier för assistans kunde påverka kraven och konsekvenserna för användaren i rörelser, inklusive muskelansträngning, kontaktkrafter i leden samt interakionskrafter mellan användaren och exoskelettet. I den femte studien analyserade vi genom simulering hur olika knäexoskelett modeller och därmed viktfördelningar påverkar användarens muskelaktivering under fyra funktionella rörelser. En knäexoskelett prototyp tillverkades och testades på fem friska försökspersoner. Exoskelettets påverkan på användarens muskelaktiveringen jämfördes i både simuleringen och i experimenten. Vi fann att muskelaktivering varierade mellan viktfördelningar och rörelser, d.v.s. ingen enstaka fysisk design var optimal för alla rörelser. Den fysiska designen för ett exoskeletett bör helst beakta användarens aktivitetsmål.

Metoderna som utvecklats i denna avhandling ger stora tillämpnings möjligheter inom reglerteknik för s.k. assistance-as-needed exoskelett.

Preface

This thesis is based on the following journal papers, which are referred to by Roman numbers. The papers are appended in full (in published or manuscript format) at the end of the thesis.

Paper 1: Longbin Zhang, Zhijun Li, Yingbai Hu, Christian Smith, Elena M. Gutierrez Farewik and Ruoli Wang. Ankle Joint Torque Estimation Using an EMG-Driven Neuromusculoskeletal Model and an Artificial Neural Network Model. *IEEE Transactions on Automation Science and Engineering 2021*; 18(2): 564-573.

Paper 2: Longbin Zhang, Xueyu Zhu, Elena M. Gutierrez Farewik, and Ruoli Wang. Ankle Joint Torque Prediction using an NMS Solver Informed-ANN Model. Submitted.

Paper 3: Longbin Zhang, Davit Soselia, Ruoli Wang, and Elena M. Gutierrez-Farewik. Lower-limb Joint Torque Prediction using LSTM Neural Network and Transfer Learning. Manuscript.

Paper 4: Longbin Zhang, Yixing Liu, Ruoli Wang, Christian Smith, and Elena M. Gutierrez-Farewik. Modelling and Simulation of a Human Knee Exoskeleton's Assistive Strategies and Interaction. *Frontiers in Neurorobotics 2021*; 15: 1-12.

Paper 5: Yixing Liu, **Longbin Zhang**, Ruoli Wang, Christian Smith, and Elena M. Gutierrez Farewik. Weight Distribution of a Knee Exoskeleton Influences Muscle activation during Movements. *IEEE Access 2021*; 9: 91614 - 91624.

Division of work between authors

The research project was initiated by Prof. Elena Gutierrez-Farewik, Dr. Ruoli Wang and Dr. Christian Smith where E.M. Gutierrez-Farewik was the main supervisor, R. Wang acted as co-supervisor and both were co-authors in Paper I, II, III, IV and V. C. Smith was co-supervisor and co-author in Paper I, Iv and V. Dr. Xueyu Zhu was advisor and co-author of Paper II. Zhijun Li and Yingbai Hu were co-authors in Paper I. Davit Soselia was co-author of Paper III. Yixing Liu was the first author of Paper V. Longbin Zhang continuously discussed the progress throughout the work with E.M. Gutierrez-Farewik and R. Wang.

Paper 1. L. Zhang designed the specific study aim. Z Li and Y Hu, constructed the preliminary ANN model and presented some preliminary results. **L. Zhang** post-processed all the data, further developed the ANN model, ran the whole framework for this study, performed statistical analysis on results, and drafted the manuscript. R. Wang collected experimental data. E.M. Gutierrez-Farewik and R. Wang supervised overall research aim and the work progress, and helped draft and revise the article. C. Smith supervised the work progress.

Paper 2. L. Zhang, X. Zhu, and R. Wang contributed to the study conception. **L. Zhang** designed the specific study aim, post-processed all the data, constructed ANN models, ran the framework for this study, performed statistical analysis on results, and drafted the manuscript. R. Wang collected experimental data. E.M. Gutierrez-Farewik, X. Zhu and R. Wang supervised the overall research aim and the work progress, and helped draft and revise the article.

Paper 3. L. Zhang designed the specific study aim, designed experiments, collected experimental data, post-processed all data. D. Soselia constructed the preliminary LSTM models and presented some preliminary results. **L. Zhang** further developed the LSTM models, designed generalization models, ran the whole framework for this study, performed statistical analysis on results, and drafted the manuscript. E.M. Gutierrez-Farewik and R. Wang supervised the overall research aim and the work progress, and helped draft and revise the article.

Paper 4. L. Zhang designed the specific study aim. Y. Liu developed the knee exoskeleton CAD model. **L. Zhang** incorporated the CAD model into a musculoskeletal model, performed the simulations and drafted the manuscript. E.M. Gutierrez-Farewik and R. Wang supervised the overall research aim and the work progress, and helped draft and revise the article. C. Smith supervised the work progress.

Paper 5. Y. Liu developed the knee exoskeleton CAD model, designed the specific study aim, performed the simulations, collected and analyzed experimental data and drafted the manuscript. **L. Zhang** incorporated the CAD model into a musculoskeletal model. E.M. Gutierrez-Farewik and R. Wang supervised the overall research aim and the work progress, and helped draft and revise the article. C. Smith supervised the work progress.

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Part I

Overview and summary

Chapter 1

Introduction

Human gaits are a repetitive sequence of limb motion that move the body forward while maintaining stance stability (Perry 1992). Neurological diseases, such as stroke, spinal cord injury, and brain trauma, often impair one's ability and strategy to walk. In recent decades, wearable exoskeletons have been developed to restore or augment motor function and/or facilitate neuro-rehabilitation. Wearable exoskeletons are promising assistive/rehabilitative devices that can help enhance motor capacity in people with strength deficits or motor disorders. The scope of the thesis is about biomechanical parameter estimation for wearable exoskeleton system design that can contribute to rehabilitation and movement assistance.

During exoskeleton-assisted rehabilitation training, patients' active participation is essential for stimulating neuromuscular recovery. However, despite recent developments in mechatronics and biosignal processing, exoskeleton control algorithms that involve the active participation of neurologically affected patients are still few. Therefore mechanistic understanding of how muscles activate to produce force and joint torque to propel the body toward a specific motion would be beneficial in the control strategy of an exoskeleton for neuro-rehabilitation, as it can guide that control to encompass and even encourage patients' active participation (Mounis et al. 2019).

While experimental evaluation of exoskeleton prototypes is important, biomechanical predictive simulations early in the design process can minimize prototype iterations and evaluate some parameters that are otherwise difficult to measure experimentally. Virtual human-machine systems (HMSs) have been used numerous times in optimal design and control of exoskeletons. However, few virtual HMSs can provide detailed impact on users at joint and muscle levels as well as interaction forces on different assistive strategies and weight distributions of exoskeletons before experiments (Shourijeh et al. 2017; Dembia et al. 2017).

1.1. Biomechanics of normal gait

When a person walks, one limb serves as a mobile source of support while the other limb advances to a new support site. Then the limbs reverse their roles.

This series of events is repeated by each limb until the person's destination is reached. A single sequence of these functions of one limb is called a gait cycle (Perry 1992). Each gait cycle is usually divided into a stance and a swing phase. As described by Perry (Perry 1992), the stance phase is the period when the foot under consideration is in contact with the ground, approximately the first 60% of the gait cycle. The stance phase can be further subdivided into four phases: (1) loading response (0-10% of the gait cycle, beginning at initial contact) is the initial double stance period and continues until the contralateral foot is lifted for swing; (2) mid-stance (10-30% of the gait cycle) is the first part of the single-leg support period that begins as the contralateral foot is lifted and continues until body weight is aligned over the forefoot; (3) terminal stance (30-50% of the gait cycle) is the second part of the single-leg support period that begins as heel rise and continues until the contralateral foot strikes the ground; and (4) pre-swing (50-60% of the gait cycle) is the second double support in the gait cycle that starts with initial contact of the contralateral foot and ends with ipsilateral toe-off. The swing phase is the rest of the gait cycle when the foot under consideration is not in contact with the ground. The swing phase can be subdivided into three phases: (1) initial swing (60-73% of the gait cycle) is approximately one-third of the swing period that begins with lift foot from ground and ends when the swinging foot is opposite the stance foot; (2) mid-swing (73-87% of the gait cycle) is the second part of the swing period that begins as the swinging foot is opposite the stance foot and ends with a vertical swinging tibia; and (3) terminal-swing (87-100% of the gait cycle) begins with a vertical tibia and ends when the foot strikes the ground.

During walking, muscles are sequentially activated in response to the stance and swing demands for weight acceptance, single limb support and limb advancement (Gage et al. 2009; Perry 1992). After initial contact, ankle dorsiflexors (tibialis anterior, extensor digitorum longus, and extensor hallucis longus) work eccentrically to decelerate ankle plantarflexion and absorb the force of impact. Ankle dorsiflexors are also active during swing to prevent foot drag. Ankle plantarflexors (principally soleus) eccentrically contract during mid-stance to control forward motion of the tibia. During terminal stance, the action of the plantarflexors (soleus and gastrocnemius primarily) switch from eccentric to concentric to facilitate heel rise and reach peak plantarflexion moment. In late pre-swing the contraction of plantarflexors reduce rapidly with the rapid decline of ground reaction force (GRF), as the loading transfers to the other limb during pre-swing (Wang 2012).

Knee extensors (quadriceps) are active in terminal swing to pre-position the foot before contact, and in the loading response to absorb shock while maintaining knee stability. Knee flexors (hamstrings primarily) act mainly to protect potential knee hyper-extension in terminal swing and early stance, and to bend the knee in initial swing to prevent foot drag. Hip extensors (primarily adductor magnus and gluteus maximus) are active during terminal swing and initial

contact to terminate and control hip flexion, and during loading response to advance the trunk over the leg. Hip flexors (primarily iliopsoas) contract to flex the hip during swing. Hip abductors (primarily gluteus maximus and gluteus medius) stabilize the hip laterally from rapid weight in early stance. Hip adductor muscles can also contribute to hip flexion in early swing (Perry 1992; Gutierrez 2003).

1.2. Pathological gait due to muscle weakness

Neurological disorders can impair a person's ability to walk, and differ markedly in their primary pathology. The abnormalities they impose on walking mechanics are characterized into four functional categories: deformity, muscle weakness, impaired control and pain (Perry 1992). In this thesis, we are mainly concerned with the pathological gait due to muscle weakness. People with muscle weakness may not be able to generate the required torques for meeting the demands of daily activities, which can adversely impact their ability to be independent and their quality of life. More than an inconvenience, muscle weakness can have a domino effect on a person's health; it is frequently followed by, for instance, slow metabolism, increase joint pressure, and postural instability (Horlings et al. 2008).

1.2.1. *Hip abductors weakness*

When there is hip abductor dysfunction, the body is less able to maintain the center of gravity on the side of the stance leg, and can lead to a drop of the pelvis towards the swing side, called a positive Trendelenburg sign. If the weakness is unilateral, the person may lean the trunk laterally toward the weak stance limb throughout the stance phase. Bilateral weakness may lead to lateral trunk lean towards the stance leg for both sides. In the long term, these compensation postures may contribute to a lateral spinal curve.

1.2.2. *Hip extensor weakness*

With hip extensor weakness, one may not be able to counteract external the hip flexor moment and may thus have a tendency for excessive hip flexion. To compensate for hip extensor weakness, one might lean the trunk posteriorly to position the GRF vector behind the hip joint center, which may over a long time lead to excessive lumbar lordosis.

1.2.3. *Hip flexor weakness*

With weakness of the hip flexors, the body may not have adequate hip flexion in early swing, potentially resulting in insufficient momentum for foot clearance and short step lengths. To compensate for hip flexor weakness, one might use the following strategies: posterior pelvis tilt, excessive knee flexion, and contralateral trunk lean.

1.2.4. *Knee extensor weakness*

If the knee extensors (quadriceps) are weak and have insufficient strength to support a flexed knee, it can be difficult or impossible to recover from any knee flexion during stance. Some strategies may be adopted to compensate for knee extensor weakness: (1) to contract hip extensors to retract the thigh and contract plantarflexors to prevent tibia advancement, resulting in knee extension or hyperextensio; (2) to lean the trunk forward to shift the GRF vector anterior to the knee, creating internal flexor moment, which may also lead to knee hyperextension; and (3) to turn the limb outward to lock the knee joint in extension. These compensations may result in a degeneration of the ligaments.

1.2.5. *Knee flexor weakness*

With knee flexor weakness, one may not be able to provide adequate knee flexion moment to restrain the thigh in terminal swing and early stance and to control the shank deceleration in the swing phase. Compensatory strategies may include increased hip flexion in the swing phase or a circumduction pattern of the swing leg.

1.2.6. *Ankle plantarflexor weakness*

With ankle plantarflexor weakness, one may not be able to counteract the external dorsiflexor moment during stance, which may result in uncontrolled tibial advancement, leading to a crouch gait position with flexed hips, flexed knees and excessive ankle dorsiflexion position during the weight bearing period. This may lead to increased activity of the quadriceps, prolonged heel contact during stance phase, short step lengths and/or slow walking speed.

1.2.7. *Ankle dorsiflexor weakness*

With ankle dorsiflexor weakness, one may be unable to counteract the external plantarflexor moment during stance and may have difficulty achieving foot clearance during swing. This may result in forefoot or wholefoot contact at initial contact and a dropfoot pattern during swing. Compensations can include increased hip and knee flexion during swing to clear the foot.

1.3. Computational methods in joint torque prediction

1.3.1. *Neuromusculoskeletal (NMS) modeling*

NMS models are physics-based mathematical representations of a person's skeletal anatomy and physiological mechanisms. Such models can help understand how muscles coordinate to generate movement and can be used to estimate joint torques during various tasks (Valero-Cuevas et al. 2009). Electromyography (EMG)-informed NMS models are commonly used to perform forward dynamics simulations of muscle dynamics and estimate joint torques using experimentally measured EMG as an input (Pizzolato et al. 2019). As described by Pizzolato

et al. (Pizzolato et al. 2015), EMG-informed NMS models usually have three neural solution modes: EMG-driven, EMG-hybrid, and EMG-assisted, based on how experimental EMG signals are used to compute muscle excitation in the NMS model.

- **EMG-driven:** Only experimentally recorded EMG signals are used to compute muscle excitations that drive the model and to estimate musculotendon forces and joint torque.
- **EMG-hybrid:** Since EMG signals of some muscles are not practical or possible to routinely collect, the muscle excitation patterns of those muscles are constructed using an optimization algorithm, such as static optimization, by minimizing the sum of squared muscle activations with known motion (Anderson and Pandy 2001). Therefore, experimentally-recorded EMG signals and muscle excitation patterns from optimization are combined to compute muscle excitation in this mode.
- **EMG-assisted:** As the quality of EMG signals may be affected by cross-talk and electrode placement, the EMG-assisted mode allows optimization to adjust muscle excitations to better track joint torque.

Among the three modes, the EMG-driven mode is adequate for joints in which EMG signals of dominant muscle are easily recorded with good quality, such as knee and ankle joints (Sartori et al. 2014; Pizzolato et al. 2015). The EMG-hybrid mode can compensate for the inability to access deep muscles such as the iliacus and psoas for hip joint motion, and the EMG-assisted mode can further compensate for the cross-talk or noisy EMGs (Hoang et al. 2018). In these situations, the EMG-hybrid and EMG-assisted models can more accurately represent the physiological properties of the NMS system; however, the two models involve an optimization algorithm to obtain the muscle activation of some muscles, which may be time-consuming and thus less suitable for real-time applications. On the other hand, the EMG-driven NMS model has been reported to robustly predict joint torque (Durandau et al. 2017), joint contact force (Pizzolato et al. 2017b) and joint stiffness (Yao et al. 2018) in real-time.

1.3.2. *Artificial neural network*

Artificial neural networks (ANNs) are increasingly popular as alternate approaches to conventional model-based techniques or as components of integrated systems. An ANN is a machine learning method. It has computational units of multiple layers that emulate the human neuronal synapse system in the brain by considering each node as artificial neurons, thereby providing the ability to process complex and non-linear information (Imran and Alsuhaibani 2019; Haeberle et al. 2019; Lee et al. 2017). An ANN usually works in two steps: first a training/learning model to map the input-output relationship, followed by execution of the trained model to estimate the output with no need to require

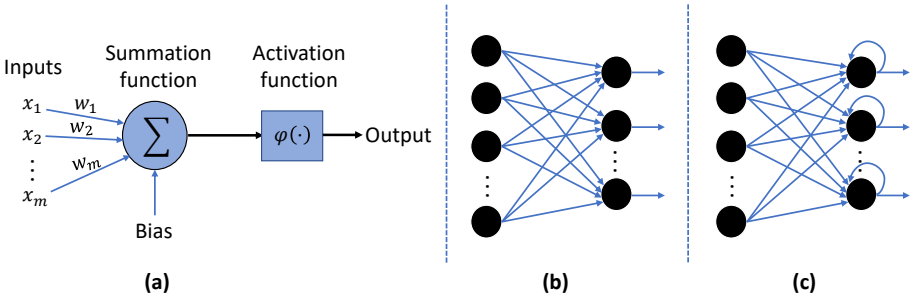


Figure 1.1: (a) The structure of one artificial neuron: it generally consists of a summation function and activation function. Summation function: each input $(x_1, x_2, ..., x_m)$ is multiplied by its respective weight $(w_1, w_2, ..., w_m)$ which measures of the input's connection strength. Weighted inputs and bias are then fed into the summing function. Activation function: Determines how much the node activates. (b) Feed-forward neural network/Multilayer perceptron: the information is only transmitted forward. (c) Recurrent neural network: similar to the feed-forward neural network structure but contains feed-back loops.

a priori mathematical knowledge between input and output. Based on the available datasets and the specific research question, different algorithms can be selected to train specific learning models. These can generally be categorized into supervised learning, unsupervised learning and semi-supervised learning. The main difference between supervised and unsupervised learning is that supervised learning uses labeled data to help predict outcomes. In supervised learning, labeled data are designed to train or "supervise" algorithms into classifying data or predicting outcomes accurately and are generally used for classification and regression problems. Unsupervised learning uses machine learning algorithms to discover hidden patterns in data without the need for human intervention, and is mainly applied for clustering, association and dimensionality reduction. Semi-supervised learning falls somewhere in between, typically using a small amount of labeled data and a large amount of unlabeled data for training. In this thesis, a supervised learning approach is used to predict joint torque values in different daily activities.

ANNs consist of multiple artificial neurons usually called nodes, that are conceptually derived from biological neurons. Each artificial neuron has inputs and produces a single output that can be sent to multiple other neurons. The most widely used neuron model is based on McCulloch and Pitts' work (McCulloch and Pitts 1943; Mellit and Kalogirou 2008)(Fig. 1.1 (a)). The artificial neuron generally consists of two parts: (1) summation function and (2) activation function. In the summation function, each input is multiplied by its respective

weight, which measures the input's connection strength. Weighted inputs and bias are then fed into the summing function. The activation function determines how much the node activates. With multiple artificial neurons interconnected to transmit information, an ANN is formed. Typically, an ANN is made of an input layer, hidden layer/s and an output layer. The input layer takes in the inputs data. The hidden layer/s receives the information from the input layer, processes it, and sent it to the output layer. The output layer computes the outputs (Imran and Alsuhaibani 2019). The interconnection of the nodes between the layers can be divided into two basic classes, namely the feed-forward neural network (FFNN)/ multilayer perceptron (MLP) and recurrent neural network (RNN). The main difference is FFNNs do not contain any feed-back loops while RNNs do (Fig. 1.1 (b) and (c)).

The MLP, the most common type of ANN, can extract features and store information on the entire network to predict very complex functions, provided enough training data is available, without constructing complex mathematically explicit formulas and costly physical models. It is thus an efficient approach to model non-linear and complex problems and has shown great power in a number of applications in biomechanics. A three-layer MLP model was used to estimate knee torques at isokinetic 30°/s and 60°/s and showed a more accurate torque prediction than a stepwise regression model (Hahn 2007). MLPs have been successfully applied to estimate ankle moments and ground reaction forces during walking and calf raises (Jacobs and Ferris 2015), elbow torque during elbow flexion (Jali et al. 2015), muscle force during dynamic contraction (Bai and Chew 2013), as well as energy expenditure during locomotion (Slade et al. 2019).

In contrast to MLPs, RNNs can model the temporal information inside a sequence, especially the relationship of the current input and former input. For this reason, it is often said that RNNs have memory. This use of "memory" makes RNNs advantageous for time series data, but introduces difficulties in model training. One problem is that it cannot store past information for very long, due to gradients vanishing or exploding. A variation of an RNN was proposed to address this problem – a long short-term memory memory (LSTM) network (Hochreiter and Schmidhuber 1997). LSTM networks can selectively remember patterns for long durations by adding memory structure that can update the information across time-steps, with gates to decide what to remember, what to forget and what to output. An LSTM network (Kim et al. 2020) was used to estimate both the wrist joint torque and position, based on EMG signals, during interaction with an environment, and estimations demonstrated high agreement with the actual values. Siu et al. (Siu et al. 2020) also successfully applied an LSTM network to estimate ankle joint torque, based on EMG and accelerometer signals during standing, walking, running, and sprinting. Both an MLP and an LSTM model were used by Mundt et al. to predict 3D lower limb joint angles and moments during gait using inertial measurement unit

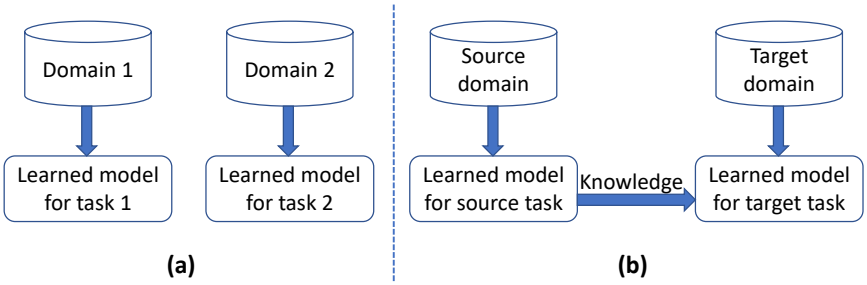


Figure 1.2: (a) Traditional artificial neural network, wherein the model learns without retaining or accumulating knowledge. (b) Transfer learning, wherein the model for a target task learns with knowledge accumulated from a previously learned model for a source task.

(IMU) data (Mundt et al. 2020). While both networks accurately predicted joint angles and moments, the MLP predicted slightly more accurately than the LSTM.

1.3.3. *Transfer learning*

Traditional ANN generally works well under a common assumption that the training and testing data have the same feature space and distribution. However, if the distribution of test data is different from training data, the prediction results are likely to degrade due to the differences. It can furthermore be difficult or expensive to recollect training data and re-train models (Pan and Yang 2009; Weiss et al. 2016; Tan et al. 2018). In such cases, transfer learning strategies have been attracting much attention to address these problems. Transfer learning is inspired by the fact that humans can learn new tasks more quickly if they have previously learned similar knowledge (Pan and Yang 2009) whereas traditional ANNs learn models independently for different tasks (Fig. 1.2). Transfer learning is generally categorized into three sub-settings: inductive, transductive and unsupervised transfer learning, based on different situations between the source and target domains and tasks. A domain $\mathcal{D}$ consists of: a feature space $\mathcal{X}$ and a marginal probability distribution $P(X)$, where $X = \{x_1, ..., x_n\} \in \mathcal{X}$. Given a specific domain $\mathcal{D} = \{\mathcal{X}, P(X)\}$, a task $\mathcal{T} = \{\mathcal{Y}, f(x)\}$ consists of two parts: a label space $\mathcal{Y}$ and an objective predictive function $f : \mathcal{X} \rightarrow \mathcal{Y}$, where the function f is used to predict the corresponding label $f(x)$ of a new instance x (Lin and Jung 2017).

- **Inductive transfer learning:** the target task is different but related to the source task, regardless of the similarities between source and target domains.

- **Transductive transfer learning:** the target task is the same as the source task wherein the target domain is different but related to the source domain.

- **Unsupervised transfer learning:** the target task and domain are both different but related to the source task and domain.

Based on "what to transfer", transfer learning approaches can be summarized into four cases in the above three sub-settings.

- **Instance transfer:** to re-weight some labeled instances in the source domain for learning in the target domain.

- **Feature representation transfer:** to find a "good" feature presentation of both source and target domains that reduces the difference between domains and the model predictive errors.

- **Parameter transfer:** to discover and transfer some shared parameters or prior knowledge between source and target domain models.

- **Relational transfer:** to build and transfer the relational knowledge mapping between the source and target domains.

Among these four, instance transfer and parameter transfer approaches are the two most popularly applied in recent research fields to solve several problems, for example, to solve limited data caused by expensive data acquisition due to ethical issues with patient data or limited research resources, and to reduce time and effort used in full re-calibration, rebuilding, or re-training for a new but related study. An instance transfer approach was incorporated into a generative adversarial network to provide enough data to train a classifier in human activity recognition systems on a target subject by transferring generated adaptive data based on source domain (subject) to the target domain (subject) (Soleimani and Nazerfard 2021). This study represented a cutting-edge solution for cross-subject knowledge transfer in human activity recognition when data acquisition and labeling are unavailable for new-users. A parameter transfer approach was adopted into LSTM neural network models to estimate skeletal muscle forces (Dao 2019), and improve prediction performance by transferring knowledge from a pre-trained LSTM model into a new model. A coupling approach of LSTM and transfer learning can potentially be applied to real-time skeletal muscle forces estimation in functional rehabilitation.

Although ANNs have been widely used in to predict joint torque based on EMG signal as inputs, and have demonstrated high predictive performance (Luh et al. 1999; Jephil et al. 2020), few studies have investigated the generalizability (ie., generalization capacity) of ANN models. Furthermore, muscle coordination patterns tend to vary across subjects and across movements (De Rugy et al. 2012; Safavynia et al. 2011). ANN models trained from multiple subjects and/or

multiple sessions are likely to result in poor performance when tested on a new subject/session, since the feature space or marginal probability distributions between training users/sessions and new user/session may be different. Collecting a large amount data on testing data on the new user/session may be expensive, time-consuming or otherwise impractical, and especially difficult for new users with motor disorders. Therefore, it is desirable to incorporate transfer learning with ANN models in joint torque prediction in order to reuse collected data from previous users/sessions to train on a new user/session.

1.4. Wearable exoskeletons

In recent decades, there has been an increasing interest in developing wearable exoskeletons to assist people with movement disorders in rehabilitation and/or in daily activities. Wearable exoskeletons are very promising assistive/rehabilitative devices that can help enhance motor capacity in people with strength deficits or motor disorders. The most concerning and difficult aspects are to control the exoskeleton to assist the users and to achieve a comfortable interface between the exoskeleton and the user.

1.4.1. *Control strategies*

During exoskeleton-assisted rehabilitation training, assistive strategies can be categorized into: passive control, triggered passive control and assist-as-needed control (Proietti et al. 2016; Marchal-Crespo and Reinkensmeyer 2009; Meng et al. 2015).

- **Passive control** refers to "exoskeleton-in-charge"; exoskeletons are driven to guide the user to follow predefined trajectories or assistive forces/torques previously extracted from healthy populations. The user only passively participates in the training exercise. This passive control technique is commonly used at early stages of therapy when an impaired limb may be unresponsive.

- **Triggered passive control** is slightly different from the passive control, as the user triggers the exoskeleton assistance. After triggering, the exoskeleton is usually controlled passively along predetermined trajectories. This technique is frequently used to introduce the brain–machine interface into the control loop to assist people with nonrecoverable impairments, such as tetraplegia (Proietti et al. 2016).

- **Assist-as-needed control** refers to "user-in-charge", "partially assistive control" and "active control"; users are dominant in performing daily tasks wherein exoskeletons provide assistance based on their intention and ability to generate torque, potentially stimulating neuroplasticity. This active control technique is usually adopted for patients with some motor capacity.

With passive control, the person is not involved in making any effort to complete tasks, preventing the exoskeletons from facilitating the activity-driven neuroplastic changes required for recovery. On the other hand, passive assistance may be needed at an early stage of therapy when motor capacity is insufficient to participate/move actively. Passive assistance early in therapy can also be useful in reducing failures, earning users' confidence and preventing negative reinforcement. As soon as the person has recovered a minimal amount of motor capacity, it is essential for neurorehabilation to involve the patient in the training process (Proietti et al. 2016).

This primary focus and context in thesis are active control techniques wherein an exoskeleton should compensate the user's existing but insufficient muscle contributions. To provide assistive torque based on the user's movement intention requires accurate and robust decoding of motor function, for instance, by recording underlying neuromuscular activities, such as brain, nerve and muscle EMG signals from the central nervous system (CNS). EMG signals are commonly used to predict the user's movement intention detection and required joint torques, as they are relatively easy to acquire and process compared to brain and nerve signals, and as they provide essential information on human motion (Sartori et al. 2018).

1.4.2. *Exoskeleton's influence on users*

While experimental evaluation of exoskeleton prototypes is crucial, biomechanical predictive simulations early in the design process can minimize prototype iterations and evaluate some parameters that are otherwise difficult to measure experimentally. Virtual HMSs have been used frequently for optimal design and control of exoskeletons. Zhou et al. (Zhou et al. 2017) created a virtual HMS by incorporating an upper limb exoskeleton into a musculoskeletal model. They evaluated two types of passive exoskeletons with gravity-compensating capability and computed optimal spring stiffness. However, different assistive strategies and their effects on the user were not studied. Pan et al. (Pan et al. 2015) simulated a novel human-exoskeleton system by integrating ADAMS and Matlab/Simulink with fuzzy proportional integral derivative (PID) algorithms. They validated the efficacy of a proposed exoskeleton control strategy in accomplishing some daily activities, such as level walking, stair ascent, and squatting down. However, they did not compute the user's effort and interactive forces. Simulation-based exoskeleton and assistive strategy design have become more common in recent years. Shourijeh et al. (Shourijeh et al. 2017) have described assistive exoskeleton torques applied to the hip, knee or ankle joint modeled as rotational actuators, and reported their effects on the overall metabolic energy expenditure. Rossi et al. (De Rossi et al. 2011), in an experimental study, measured the contact pressure distribution in a user wearing a lower limb exoskeleton, using an array of soft silicone pressure sensors. This type of study is useful in assessing the safety and comfort of human-machine interactions, but methodological limitations in experimental studies make it impossible to

study other important information in the human-machine interaction, such as joint contact force and the effects of different assistive strategies. In addition, limited research has been conducted on how different weight distributions of knee exoskeleton components can affect efficacy or comfort. Therefore, it is useful to create a virtual HMS to study how different assistive strategies and weight distributions of exoskeletons affect users before conducting experiments.

Chapter 2

Specific aims

The scope of the thesis is about biomechanical parameter estimation for wearable exoskeleton system design. This thesis is based on five studies that focus on different aspects of exoskeleton controller design and physical design, within a context of contributing to the design of wearable exoskeleton systems for rehabilitation and movement assistance. Our goals were to robustly and accurately estimate a user's physiological input, i.e. generated joint torque in real-time using wearable sensors, and to develop a framework to study how different control strategies and physical designs may influence the user.

Study I: *Ankle Joint Torque Estimation using an EMG-Driven Neuromusculoskeletal Model and an Artificial Neural Network Model*

Motivation: Both EMG-driven NMS and ANN model-based methods have been applied frequently in limb joint torque estimation, but their respective advantages in prediction have not been well investigated.

Objective: To estimate ankle joint torque using an EMG-driven NMS model and a standard-ANN model simultaneously.

Study II: *Ankle Joint Torque Prediction using an NMS Solver-Informed ANN Model*

Motivation: Results from Study I suggested that the standard ANN model could predict ankle torques more accurately than NMS models when trained on a large and varied set of trials, whereas the NMS model in ankle torque prediction was more accurate for movements that have similar muscle contraction coordination as in the calibration movements and possibly more useful due to its descriptions of underlying physical principles. A logical question that arises is whether it would be beneficial to train ANN models in combination with physics-based models.

Objective: To estimate ankle joint torque by an NMS solver-informed ANN model (hybrid-ANN) and investigate the generalizability of the hybrid-ANN model across subjects.

Study III: *Lower-limb Joint Torque Prediction using LSTM Neural Network and Transfer Learning*

Motivation: Few studies describe torque prediction in multiple lower-limb joints and degrees of freedom, nor evaluate the generalizability of LSTM networks across subjects and daily tasks.

Objective: (1) To predict hip flexor/extensor (F/E), hip abductor/adductor (Ab/Ad), knee flexion/extension and ankle dorsiflexor/plantarflexor (D/P) torques using LSTM neural networks trained with EMG and kinematics from single and from multiple movements; (2) to evaluate the generalizability of the LSTM models across subjects and tasks, and (3) to implement a transfer learning strategy to further improve the model prediction performance by making use of the knowledge learned from previous experience.

Study IV: *Modeling and Simulation of a Human Knee Exoskeleton's Assistive Strategies and Interaction*

Motivation: While experimental evaluation of exoskeleton prototypes is important, biomechanical predictive simulations early in the design process can minimize prototype iterations and evaluate some parameters that are otherwise difficult to measure experimentally. The exoskeletons' influences on the user at joint and muscle levels, including muscle activation, physiological joint torque, joint contact forces and interaction forces have not been extensively described.

Objective: To create a simulation of a virtual HMS with a knee exoskeleton, to evaluate how different assistive strategies affect the user's required effort as well as joint contact forces, and to study the interaction forces between the device and the user.

Study V: *Weight Distribution of a Knee Exoskeleton Influences Muscle activation during Movements*

Motivation: Lower extremity powered exoskeletons help people with movement disorders to perform daily activities and are increasingly used in gait re-training and rehabilitation. Studies of powered exoskeletons often focus on technological aspects such as actuators, control methods, energy and effects on gait. Limited research has been conducted on how their mechanical design parameters can affect efficacy or comfort.

Objective: To determine how different mechanical designs of a knee exoskeleton influence the user's required effort during different movements, using a combination of simulation and experimentation.

Chapter 3

Methods

3.1. Subjects

In **Studies I and II**, a convenience sample of ten able-bodied subjects (age: 26 ± 2.9 years; weight: 70.4 ± 11.5 kg; height: 175.1 ± 8.5 cm; sex: 4F/6M) were recruited among students and colleagues. In **Study III**, a convenience sample of eight able-bodied subjects (age: 29 ± 4 years; weight: 65.2 ± 17.8 kg; height: 168.1 ± 9.4 cm; sex: 4F/4M) were recruited. In **Study V**, five able-bodied subjects (age: 25 ± 6.0 years, weight: 53.6 ± 7.4 kg, height: 165.2 ± 9.2 cm, sex: 3F/2M) were recruited to participant in validation experiments. All subjects participated voluntarily and gave informed written consent. All studies were approved by the Swedish Ethical Review Authority (Dnr. 2016/286-32 and 2020-02311).

3.2. Experimental procedures

In **Studies I and II**, subjects were asked to perform seven movements: walk with different speeds (slow, self-selected speed, and fast), isokinetic ankle dorsiflexion and isokinetic ankle plantarflexion (two different speeds at 60°/s and 90°/s). In **Study III**, subjects were asked to walk quickly, walk at a self-selected speed, walk slowly, jump down from a stair, jump up to a stair, jump up, land, squat, stand up from a seated position (sit-to-stand) and sit down (stand-to-sit), in a randomized sequence, and to repeat each test 12-15 times. In **Study V**, subjects were asked to walk at a comfortable speed and to squat from a standing position, both while donning a knee exoskeleton in unpowered mode, and without the exoskeleton. Each movement was repeated at least ten times in each condition. In **Studies I and II**, surface EMG signals (Noraxon Inc., AZ, USA) of tibialis anterior (TA), soleus (SOL), and gastrocnemius medialis (GM) of each subject's right leg were recorded at 3000 Hz and electrodes were placed according to European recommendations for surface EMGs (Hermens et al. 1999). In **Study III**, surface EMG signals (Myon/Cometa aktos, Italy) of SOL, TA, GM, peroneus longus (PL), gastrocnemius lateralis (GL), rectus femoris (RF), vastus medialis (VM), vastus lateralis (VL), biceps femoris (BF), semitendinosus (ST), gluteus maximus (GLMA), gluteus medius (GLME), and tensor fasciae latae (TFL) muscles were recorded at 1000 Hz. The electrodes were placed unilaterally on each subject. A randomization procedure was used

to determine on which leg EMG electrodes were to be placed. In **Study V**, surface EMG signals (Myon/Cometa aktos, Italy) of RF, VM, VL, ST, BF, SOL, GL, GM, and TA of the right leg were recorded at 1000 Hz. Each subject's maximum voluntary isometric contraction (MVIC) was measured to normalize of each subject's EMG data (Konrad 2005).

In Studies I and II, each participant was encouraged to plantarflex and dorsiflex with maximum effort within a range of motion of 15° dorsiflexion and 20° plantarflexion, five consecutive times in isokinetic movement tasks. The motions were recorded by a dynamometer (IsoMed 2000, Hemau, Germany) at 5000 Hz. Each participant walked for at least three recorded trials at a self-selected speed, then at 100 steps/min, and 120 steps/min timed with a metronome, respectively. Motion were recorded in a 3D motion capture system (Qualisys, Gothenburg, Sweden) at 200 Hz by tracking the reflective markers placed on the participants based on a Plug-in-Gait marker set protocol (Kadaba et al. 1990; Davis III et al. 1991). Ground reaction forces (GRFs) were measured by two force plates (3000 Hz, Kistler, Winterthur, Switzerland) at 3000 Hz. **In Studies III and V**, motions was recorded in a 3D motion capture system (Vicon V16, Oxford, UK) at 100 Hz by recording the marker trajectories placed on the subjects according to the CGM2.3 marker set protocol (Leboeuf et al. 2019) for **Study III** and the Plug-in-Gait marker set protocol for **Study V**. GRFs were measured by three force plates (AMTI, MA, USA) at 1000 Hz.

3.3. Data processing (Studies I, II and III)

EMG signals were processed by band-pass filtering (30-300 Hz), rectifying, low-pass filtering (6 Hz), and normalizing to the maximum EMG value obtained from processed EMGs across all trials (Pizzolato et al. 2017a; Sartori et al. 2016; Hoang et al. 2018).

Joint angles were computed using an inverse kinematics method in which a weighted least square problem that minimizes the distance between modeled and measured markers is solved. Joint torques were computed through the inverse dynamics method by solving the dynamic equation of motion (Eq. (3.1)) (Zajac et al. 2002)

$$\mathbf{M}(q)\ddot{q} + \mathbf{C}(q,\dot{q}) + \mathbf{G}(q) = \tau, \tag{3.1}$$

where $q, \dot{q}, \ddot{q}$ are the position, velocity, and acceleration of the generalized coordinates, respectively. $\mathbf{M}(q)$ is the mass matrix, $\mathbf{C}(q,\dot{q})$ is the centripetal and Coriolis forces matrix, $\mathbf{G}(q)$ is the gravitational forces matrix, and τ is the vector of unknown generalized forces.

3.4. Joint torque prediction models

3.4.1. *EMG-driven neuromusculoskeletal model (Studies I and II)*

We used an open-source EMG-informed NMS model developed by Pizzolato et al. (Pizzolato et al. 2015) consisting of four elements (see Fig. 2 in Paper

I): muscle activation dynamics, muscle contraction dynamics, musculotendon kinematics, and joint dynamics (Sartori et al. 2011).

The muscle activation dynamics element is used to compute muscle activation with acquired EMG signals as input. The acquired EMG signal $e(t)$ is related to neural activation $u(t)$ in a time domain t, as Eq. (3.2) (Lloyd and Besier 2003).

$$u(t) = \alpha e(t-d) - \beta_1 u(t-1) - \beta_2 u(t-2), \tag{3.2}$$

where α is the muscle gain coefficient; d is the electromechanical delay; and β_1 and β_2 are the recursive coefficients. The neural activation is related to muscle activation $a(t)$ and the relationship is formulated as Eq. (3.3):

$$a(t) = \frac{e^{Au(t)} - 1}{e^{A} - 1}, \tag{3.3}$$

where A is the non-linear shape factor (Buchanan et al. 2004; Hoang et al. 2018).

The muscle contraction dynamics element is used to calculate musculotendon forces using a Hill-type muscle model. The relationship between force in each musculotendon unit (MTU) F^{mt}, muscle activation and kinematics is defined as Eq. (3.4),

$$F^{mt} = F_0^m [f_a(\tilde{l}_m) \cdot f_v(\tilde{v}_m) \cdot a + f_p(\tilde{l}_m) + d_m \tilde{v}_m] cos(\phi), \tag{3.4}$$

where F_0^m is the maximum isometric muscle force, $\tilde{l}_m$ is the fiber length, $f_a(\tilde{l}_m)$ describes the active force-length relationship, $\tilde{v}_m$ is fiber contraction velocity, $f_v(\tilde{v}_m)$ describes the force-velocity relationship, $f_p(\tilde{l}_m)$ is the passive force-length relationship, ϕ is the fiber pennation angle, and d_m is the muscle damping parameter.

The musculotendon kinematics element is used to calculate moment arms and musculotendon lengths of each MTU using a musculoskeletal geometric model. Finally, the joint dynamics element is used to predict joint torques through the product of musculotendon forces F_{mt} and their moment arms r_{mt}.

3.4.2. *Artificial neural network models (Studies I, II and III)*

3.4.2.1. *Feed-forward ANN/MLP models (Studies I and II)*

We constructed a standard feed-forward ANN/MLP model to predict ankle joint torque in Study I with the same inputs that we used in the EMG-driven NMS model, specifically EMG signals and joint angle (see Fig. 3 in Study I). Since the standard feed-forward ANN technique is a purely data-driven method and would become less reliable when encountering unseen movements, we constructed a hybrid-ANN model in Study II that leverages more informative physical features, specifically ankle joint torque and muscle forces computed by the NMS solver (see Fig. 3 in Study II).

The feed-forward ANN models consist of an input neural layer, n hidden layers and an output neural layer. The standard-ANN model included four neurons in the input layer, which are experimentally measured EMG data from three muscles and ankle joint angle. Besides these input variables, hybrid-ANN added four additional input features: ankle joint torque and three muscle forces computed by the NMS solver. For the intra-subject scenario, two hidden layers were used; four hidden layers were used in the inter-subject scenario. In the output layer, the predicted ankle joint torque was determined. The mean square error between predicted and experimentally measured ankle joint torque is used during the training process. The hyper-parameters of the ANN models were determined by "coarse-to-fine" random search (Bergstra and Bengio 2012).

3.4.2.2. *Long short-term memory models (Study III)*

We constructed LSTM neural networks to predict joint torque based on EMG signals and joint angles (see Fig. 2 in Study III). The EMG signals and joint angles were acquired from 3D motion analysis and pre-processed as input data for the LSTM neural networks. For each LSTM neural network, the time-series input data were transformed as time slices data. Each time slice included five time steps, and each time step had seventeen features including thirteen muscle EMG signals and four joint angles. n LSTM layers ($n = 2$ for single-task and $n = 3$ for multiple-task in the intra-subject scenario; $n = 3$ for single-task and $n = 4$ for multiple-task in the inter-subject scenario were included, followed by a dense layer and a dropout layer. The joint torque was predicted at the output layer. The hyper-parameters of the LSTM models were also determined by "coarse-to-fine" random search.

3.5. Transfer learning (Studies II and III)

Generally, trained models in inter-task/subject scenarios would face a performance drop when tested with unseen data from new tasks and/or users (Soleimani and Nazerfard 2021; Su and M Gutierrez-Farewik 2020). To boost the prediction performance, transfer learning methods (see Fig. 4 in Study II and Fig. 3 in Study III) are used in the inter-task/inter-subject scenarios to learn structural similarities by pre-training a model on multiple subjects/tasks and transfer the learned knowledge to a target task/subject. In the pre-trained model, we extracted the layers except for last several layers and transferred them to the target model. In the target model, we added new layers, and then trained the weights of new layers and fine-tuned the weights of other layers with the data from the target task/subject. The fine-tuning technique is an optional step in transfer learning and can potentially achieve meaningful improvements by incrementally adapting the pre-trained features to the new data (Shin et al. 2016).

3.6. Human-machine system (Studies IV and V)

We created a human knee exoskeleton CAD model, then incorporated it into a musculoskeletal model (AnyBody AMMR Twente Lower Extremity Model Version 2.0) (Carbone et al. 2015), creating a virtual HMS.

The CAD (Solidworks) model was made based on a physical knee exoskeleton, consisting of a Brushless DC motor (EC 90 flat 90 Watt, Maxon, Switzerland), a gear reducer (Planetary Gearhead GP 52C, Maxon, Switzerland), a bevel gear to change the direction of applied torque, thigh and shank cuffs, knee joint connection, and strap bases. The varied assistive torque generated by the exoskeleton was modeled as an ideal torque. Mass, moment of inertia and damping of the exoskeleton components were incorporated into the virtual HMS.

In the virtual HMS, three different assistive strategies, namely a rotational actuator (RA), a simple pendulum model (SPM) and a damped pendulum model (DPM), were used to simulate assistive torques at the knee during normal and fast gait. Outcome measures including muscle impulse, required physiological knee torque, joint contact forces and interaction forces were estimated through inverse dynamic analysis, with muscle recruitment and activation resolved through static optimization with an assumption of minimizing the sum of the squared muscle activation (Damsgaard et al. 2006; Bassani et al. 2017; Rasmussen et al. 2001).

Four knee exoskeleton CAD models were also developed in the virtual HMS to study the effects of weight distributions of the knee exoskeleton components on simulated muscle activities during three functional movements, i.e., normal gait, squat, and ramp ascent (8°).

3.7. Data Analysis

3.7.1. *Studies I, II and III*

In Study I, to investigate and compare the accuracy of predicting ankle torque through an NMS and a standard ANN model, six different cases were formed, using different types of trials to calibrate (NMS) or train (ANN) the model in the intra-subject scenario. Calibrated (NMS) and trained (ANN) models were then tested on the remaining trials. In Study II, the ankle torque prediction performance of NMS, standard-ANN and hybrid-ANN models were further investigated in two scenarios: intra-subject and inter-subject. The benefit of adding transfer learning was also investigated in the inter-subject scenario (Fig. 5 in Study II). For each scenario, seven different cases were studied, using different collected movement types and speeds to calibrate (NMS model) or train (ANN models) the models. In Study III, performance of LSTM models to predict hip F/E, hip Ab/Ad, knee F/E and ankle D/P joint torques was investigated in two scenarios—intra-subject and inter-subject scenarios. Each scenario was further divided into intra-task and inter-task tests. In addition,

the benefit of adding transfer learning was also investigated across tasks and subjects (Fig. 4 in Study III). For each test, different cases were studied using different collected movements to train the models.

The accuracy (Studies I, II and III) of joint torque prediction models was defined as low root mean square error (RMSE) between measured (computed through inverse dynamics) and estimated joint torque, normalized by body mass. RMSE was computed for each subject and model, and then averaged across the subjects. Paired t-tests (Study I) and Wilcoxon signed-rank tests (Studies II and III) were used to investigate whether the RMSEs computed from the various models differed ($p < 0.05$ significance level). In Study III, normalized root mean square error (NRMSE) was also computed as RMSE divided by the range of measured joint torque during different movements and used for further analysis.

3.7.2. *Study IV*

Gait at a normal velocity of $1.22m/s$ and at a fast velocity of $1.49m/s$ was simulated, using kinematics available in the software, with three different assistive modes for the exoskeleton knee joint — RA-based, SPM-based and DPM-based assistance — as well as a no-assistance (NA) mode with added dead weight from the exoskeleton's components. Muscle impulse, defined as the integral of muscle force with respect to time during one gait cycle, was computed for knee flexors: sartorius (SAR), BF, ST, semimembranosus (SM), gracilis (GRA), GAS, and for knee extensors: VL, VM, RF, vastus intermedius (VI).

3.7.3. *Study V*

Gait, squat, and ramp ascent were simulated for each exoskeleton physical design for each subject. Recorded EMG signals during gait and squat movements were band-pass filtered (Butterworth 20 Hz to 400 Hz), rectified, then low-pass filtered (Butterworth 4 Hz), and normalized to each muscle's MVIC. Simulated and measured muscle effort, defined as integrated muscle activities during each movement cycle, were compared. The changes in muscle activities/excitation ratios of donning a knee exoskeleton to those without wearing a knee exoskeleton were computed and compared.

The normality of data distribution was evaluated by a Shapiro-Wilk test for simulated and experimental results. Paired t-tests for data that was normally distributed and Wilcoxon rank-sum tests for data that was not normally distributed were used to determine whether differences were observed between the changes in ratios from integrated muscle activities and those from integrated muscle excitation in both gait and squat movements.

Chapter 4

Results and Discussion

4.1. Joint torque estimation (Studies I, II and III)

4.1.1. *NMS vs. standard ANN model (Study I)*

The main contribution of this study is that we presented detailed comparisons of ankle joint torque prediction between an NMS model and a standard ANN model (Table 4.1). The detailed analysis of the methods' performance in predicting ankle joint torque can significantly contribute to determining which model to choose, and under which circumstances, and may thus be of large benefit for exoskeleton controller design. Few studies have compared NMS- and ANN-based approaches to predict joint torques during daily activities. This is the first study comparing joint torque prediction performance of an EMG-driven NMS model to a standard ANN in different movements, using different training data input.

Table 4.1: Results summary in Study I: ankle dorsiflexor/plantarflexor (D/P) torque prediction by NMS and standard ANN models in intra-subject scenario.

Torque prediction	NMS vs. standard ANN in intra-subject scenario
Ankle D/P	Standard ANN was less accurate for unseen movements;
	Standard ANN was more accurate when trained on a large and varied set of trials;
	NMS was more accurate for movements that have similar muscle contraction coordination as in the calibration movements

The standard ANN prediction performed well when trained on data from similar movements but was less accurate when predicting torques in unseen movements (in Case 1, 2 3 and 4, see Fig. 4 in Study I). This finding is consistent with the statement in a recent review of predictive simulation methods for human gait analysis by Ezati et al. (Ezati et al. 2019); they discuss the limitation of neural network that the trained NN model might not be accurate when the

collected experimental data is not sufficient. Calibrating on a large number of trials improved the performance of the standard ANN, but did not in general improve the performance of the NMS in predicting joint torques (in Case 5, see Fig. 4 in Study I). However, the trained EMG-driven NMS model predicted ankle torques more accurately than the standard ANN, in movements that have similar muscle contraction coordination as in the calibration movements (in Case 2, see Fig. 4 in Study I).

Our findings suggested that the NMS model was overall more useful in ankle torque prediction as it uses domain knowledge to reproduce the transformations from EMG signals and joint angles to musculotendon forces and joint torques. In contrast, the standard ANN model could predict ankle torques more accurately when trained on a large and varied set of trials, but less accurately in unseen movements as it only learns the relationships between inputs and outputs in the trained movements, without any consideration to the underlying physical relationships.

Table 4.2: Results summary in Study II: ankle dorsiflexor/plantarflexor (D/P) torque prediction by NMS, standard-ANN and hybrid-ANN models.

Torque prediction	NMS vs. standard-ANN vs. hybrid-ANN in intra-subject scenario	NMS vs. standard-ANN vs. hybrid-ANN in inter-subject scenario without transfer learning	NMS vs. standard-ANN vs. hybrid-ANN in inter-subject scenario with transfer learning
Ankle D/P	Hybrid-ANN model was most accurate;	Both standard- and hybrid-ANN models predicted torque poorly;	Torques predicted by standard- and hybrid-ANN were significantly more accurate in all cases;
	Torques predicted with NMS model have small offsets at the beginning of the cycle;	Torque prediction by standard-ANN was less accurate than by hybrid-ANN	
	Hybrid-ANN predicted torque poorly in one subject for Case 2 in tested movements		Prediction accuracy when the hybrid-ANN model performed poorly in one subject in Case 2 was improved

4.1.2. *A combined NMS and standard-ANN model (Study II)*

The main contribution of this study is that we proposed an NMS solver-informed ANN model to predict ankle joint torques, leveraging the advantages of both models. This study investigated the joint torque performance of three models

in two scenarios: intra-subject and inter-subject tests (Table 4.2). In the inter-subject tests, we adopted a transfer learning technique to further improve the prediction accuracy by taking advantage of the information extracted from previous subjects. To the best of our knowledge, this study represents the first attempt to apply an NMS solver-informed ANN model to predict joint torques, and the first attempt to adopt a transfer learning approach into ANN models to predict torques in different/unseen subjects during daily activities.

Overall, the hybrid-ANN model predicted ankle joint torques more accurately than both the NMS and the standard-ANN model, presumably attributable to the additional physical features obtained by the NMS model. The important physical features, i.e., individual muscle forces and joint torque, extracted from the NMS model into the ANN model design, presumably improved the model's performance in predicting joint torque; individual muscle forces are the dominant intermediate components in joint torque computation based on the joint dynamics model (Fig. 2 in Study II) and the joint torques from the NMS-solver were able to provide more relevant physiological features to train the hybrid-ANN model. These features can complement standard ANN models that might become less accurate when encountering unseen movements. For example, compared to hybrid-ANN and NMS models, the lowest torque prediction accuracy was observed in the standard-ANN for unseen tested movements— for instance, slow walking in Case 2 in the intra-subject test (Fig. 6 H_2 in Study II).

The torques predicted by the NMS model displayed small offsets at the beginning of the cycle, for example, G_2 in Fig 6, and A_2 and C_2 in Fig 7 of Study II, show some offsets at the beginning of the cycle when the measured torque was close to zero. This may be attributable to that fact that the two previous time-steps of muscle-tendon unit neural activation were needed to compute muscle neural activation (Eq. (3.2)), but at the beginning of the cycle, any earlier neural activation values were not available. Instead, they were approximated using EMG signals from two previous time-steps and may thus have resulted in the observed offset in predicted torque. This offset was eliminated in the hybrid-ANN, which can be regarded as the correction mapping from joint torque predicted by the NMS model to the measured torque. That is possible because, compared to NMS, the hybrid-ANN model has one more step: adding neural networks to better train a model by minimizing the errors between measured and predicted joint torques.

Our findings suggest that the NMS solver-informed ANN model predicted ankle joint torque more accurately than both the NMS and the standard ANN models, and indicate a potential benefit of integrating features of the NMS model into standard ANN models.

It is important to note that the hybrid-ANN predicted torques with poor accuracy in one subject for Case 2 in tested movements: fast walking and slow walking (Fig. 6 B_1 and H_1 in Study II) in the intra-subject tests. This may be due to several reasons. First, the hybrid-ANN model adopted NMS-solver informed muscle forces and joint torque as input features. However, in certain situations, the torques predicted by the NMS were inaccurate, and may have provided less informative or even misleading input features to the hybrid-ANN model. Second, we have already observed that the NMS model performed torque with poor accuracy for some tested movements if the muscle coordination patterns differed greatly between the calibration and the testing movements. In future work, we suggest the use of a Bayesian neural network that offers uncertainty quantification of the prediction that would be adopted to auto-detect such situations first and therefore guide the enrichment of training data to re-train a robust model (Kwon et al. 2020).

Table 4.3: Results summary in Study III: hip flexor/extensor (F/E), hip abductor/adductor (Ab/Ad), knee flexor/extensor and ankle dorsiflexor/plantarflexor (D/P) torque prediction using LSTM neural networks and transfer learning.

Torque prediction	One uniform model vs. ten separate models in intra-task tests in intra-subject scenario	One uniform model in inter-task tests with vs. without transfer learning in intra-subject scenario	One uniform model and ten separate models with vs. without transfer learning in inter-subject scenario
Hip F/E, hip Ab/Ad, knee F/E and ankle D/P	Both models predicted multiple joint torques accurately;	Uniform model predicted torque innaccurately when transfer learning was not included; With transfer learning, torque prediction accuracy was significantly improved	Without transfer learning, both the uniform model and ten separate models predicted joint torque less accurately than in intra-subject scenarios; With transfer learning, torque prediction accuracy was significantly improved. Notably, joint torques were accurately predicted with only a few movements from "new" subjects.

4.1.3. *LSTM model (Study III)*

The main contribution and novelty of this study was analysis of LSTM model robustness in predicting joint torques several joints and planes during daily activities using LSTM neural networks and transfer learning (Table 4.3) and the evaluation of the generalizability of LSTM networks to predict torques

across subjects and across movements/"tasks". In this study, ankle and knee joint torques in the sagittal plane, as well as hip joint torques in both sagittal and frontal planes during several different motions were predicted using LSTM neural network models in intra- and inter-subject scenarios, and in intra- and inter-task tests. We constructed different LSTM networks in different cases to predict joint torques, and investigated their generalizability across movements and subjects. Our findings, particularly those relating to how to create and train generalizable models with inputs from few movements, can provide useful guidelines for training models when only minimal amounts of movement input for training are feasible. This model thus holds great promise in applications of assistance-as-needed exoskeleton control design, in which the intended users may not be able to undergo an extensive battery of movements to train the model.

The LSTM models predicted multiple joint torques accurately both through a uniform model trained by multiple-tasks and through ten separate models trained by a single task each in intra-subject tests (Case 1 and Case 2 in Figure 5 in Study III). Both of these models were tested with the unseen, "new" trials of the same movement types used for training. Generally, artificial neural networks can be expected to predict accurately when trained and tested on the same motions. The constructed LSTM model with multiple layers and neurons, as one type of artificial neural networks, is capable of selectively remembering patterns for a duration of time, and could extract the complex nonlinear relationship between EMG signals and joint torques, and thus accurately predict torques in unseen trials of the same movement types, regardless of whether the network consists of a uniform model or of ten separate models.

Our results illustrate that both a uniform model trained on all ten movements and ten separate LSTM models trained on one movement each could predict lower limb joint torques accurately, i.e. with low prediction error.

4.1.4. *Transfer learning (Studies II and III)*

Transfer learning techniques were adopted to improve the joint torque prediction accuracy across subjects in Studies II and III and across tasks in Study III (Table 4.2 and 4.3). Transfer learning is a popular technique in cross-subject or cross-task validation, and can prevent the performance drop mentioned earlier by adapting obtained knowledge from the source domain (previous subjects/tasks) to the target domain (new subject/task) (Soleimani and Nazerfard 2021). In Study II, joint torque prediction was significantly more accurate in all cases after adopting transfer learning. It is important to note that the prediction accuracy when the hybrid-ANN model performed poorly in one subject in Case 2 was improved after adopting transfer learning and even better than prediction from the intra-subject test (Fig. 6 B_1 and H_1 VS Fig. 12 B_1 and H_1 in Study II). In Study III, the generalizability of the LSTM models was poor in inter-task tests

of intra-subject scenarios (NRMSE $\leq$ 18.4%) and in intra-task tests of inter-subject scenarios (NRMSE $\leq$ 21%) when transfer learning was not included. Torque prediction accuracy was significantly improved in all cases when transfer learning technique was included. For instance, the maximum prediction errors in squat in inter-task (NRMSE 18.4%) and inter-subject (21.0%) scenarios reduced (8.7% and 9.6%, respectively) with transfer learning. It is important to note that the generalized model could predict joint torques with similar accuracy in inter-subject Case 7 (re-trained on all movements) and Case 9 (re-trained on a subset of 5 movements), indicating that the model's generalizability when trained on fewer movement types is not compromised. This possibly benefits from the shared similarities in the pre-trained model based on previous subjects, which was used as the starting point to train a new model for the new user. The starting point would influence the performance of the trained neural network model wherein the closer initial weights are close to the target solution, the better performance of the network achieves (Dao 2019; Hinton and Salakhutdinov 2006). In short, when transfer learning was incorporated in inter-subject/task scenarios, LSTM models predicted joint torques accurately with little effort from "new" subjects. This type of model may thus be particularly useful for applications in which acquiring a large amount of training data may not be possible.

4.2. Human-machine system (Studies IV and V)

4.2.1. *Evaluation of control strategies influence on the user (Study IV)*

The main contribution of this study was the development of a virtual HMS that fully incorporates both a knee exoskeleton hardware and control into a detailed musculoskeletal model, and its demonstrated use in computing how different assistive strategies may affect the user's required torques, muscle impulses, contact forces and interactive forces during movement. Detailed information at joint and muscle levels provided by using a musculoskeletal modeling environment can be important factors in optimal controller design of exoskeletons.

To illustrate the potential use of the developed virtual HMS, we focused on the knee in normal gait, and adopted three common but conceptually different assistive strategies at the knee joint (RA, SPM and DPM strategies) to determine how they may influence the user. We found that compared to no assistance, assistance torque based on DPM models can reduce the demand on the user's knee flexors while the RA-based model reduced the demand on knee extensors by a small amount (Fig. 5 in Study IV). This finding illustrates that the knee exoskeleton with the DPM-based assistive strategy could, for instance, assist a person with weakness in the knee flexors to achieve normal and even fast walking kinematics, despite the weakness.

Despite the lower demand on the user's muscles in SPM and DPM assistance modes than in other modes, it is notable that the interactive forces between

the user and the exoskeleton straps were higher in the pendulum-based models in late stance and late swing (Fig. 7 in Study IV). In addition, we found the highest compressive tibiofemoral force and sartorius and hamstrings impulses during late stance and late swing in the no-assistance mode (Fig. 6 in Study IV). The comparatively lower compressive tibiofemoral force in the SPM and DPM modes can be in part attributed to the reduced muscle impulse during late stance and late swing. High compressive tibiofemoral force has frequently been used as an indicator in gait analysis (Mannisi et al. 2019; Lee and Wang 2015; Wang et al. 2016) associated with knee pain.

The results of the illustrative example suggest that both joint position and rotational direction should be considered in an assistive strategy, and that a damping component in a pendulum-based assistive strategy could be beneficial to further reduce the user's required muscle effort during the stance phase. We have also demonstrated that a higher assistive torque may incur a trade-off in the form of higher interaction forces, which should be kept in mind when designing a control strategy.

4.2.2. *Different weight distributions of a knee exoskeleton: effect on users' muscle activation (Study V)*

Studies of powered exoskeletons often focus on technological aspects such as actuators, control methods, energy and effects on gait. However, limited research has been conducted on how different mechanical design parameters can affect the user. The main contribution of this study was the analyses of how different mechanical designs of a knee exoskeleton influence the user's required effort during different movements, using a combination of simulation and experimentation. In a simulation with a virtual HMS, four realistic knee exoskeleton CAD models were developed and their effects on simulated muscle activities were studied during three functional movements. The detailed information of how a powered exoskeleton's core components may affect a user's ability and performance can contribute to exoskeletons' physical design for different daily activities.

We have shown through simulations that exoskeleton physical designs and weight distributions influence muscle activities in different ways during different movements. No single physical design was optimal for all movements. Effects of an exoskeleton's distribution of bulky components are not negligible. Furthermore, an exoskeleton did not always induce more muscle activities than without, which indicates that muscle synergy patterns could be altered with the exoskeleton, and that its weight can be advantageous in some instances/movements. In addition, the robustness of our simulation was tested by comparing their agreement with experimental results of subjects donning one of the physical designs while performing two of the simulated motions.

Our findings reveal that exoskeleton weight distribution can influence muscle activities, and that the influence is movement-dependent. No single physical design was optimal for all movements. A user's daily routine, activity goals and residual muscle strength should be considered while designing an exoskeleton's physical design and control algorithm.

4.3. Limitations

There are of course some limitations in this thesis. We only constructed one NMS model to predict joint torque. In addition, a subset sensitivity analysis of some parameters in the NMS model, such as electromechanical delay, muscle damping coefficient and tendon types, can be further investigated to determine the optimal parameters for joint torque prediction. Only MLP and LSTM neural networks were investigated; more complicated architectures, such as convolutions neural networks, can be further investigated for joint torque prediction. The performance of using different neural networks can be further compared and analyzed. In the virtual HMS, only three control strategies of exoskeleton's effect on the user was analyzed during gait; it should be technically possible to analyze a greater number of control strategies in the HMS. The strap's tension, material, and surface friction were not considered in the HMS, nor were the soft tissue properties of the user's thigh and shank, which will all affect the computed interaction forces. In the experimental movements, one could consider standardizing the chair height in relation to each subject's height during sit-to-stand and stand-to-sit, and could also take into account different subjects' squat techniques, as they can result in different muscle activation patterns. Also, this study only includes studies of able-bodied persons, even though the context in focus, exoskeleton users, will largely consist of patient populations. Furthermore, the able-bodied subjects enrolled in this thesis were relatively homogeneous with regards to age and overall fitness.

Chapter 5

Conclusions and future work

This thesis presented studies in which we robustly and accurately estimate joint torque using machine learning models and neuromusculoskeletal models in daily activities. The detailed analysis of different models' performance in predicting joint torques can contribute to determining which model to choose, and under which circumstances, and thus be of large benefit for exoskeleton rehabilitation controller design. This thesis also analyzed different assistive strategies of an exoskeleton's influence on users as well as the effects of different weight distributions of a exoskeleton on users, which makes it possible to estimate biomechanical parameters early in the design process. This can ultimately minimize prototype iterations and evaluate parameters that are otherwise difficult to measure experimentally.

The standard ANN/MLP model predicted ankle torques somewhat better when trained on a large and varied set of trials but less accurately in unseen movements whereas the NMS model was overall more useful due to the underlying physical principles. The combination model of NMS and standard-ANN, i.e., NMS solver informed-ANN model, had a better torque prediction performance than both the NMS and the standard ANN models, indicating the potential benefit of integrating physical features from NMS model into standard ANN models. Furthermore, constructed LSTM models illustrated both a uniform model trained on several movements and several separate LSTM models trained on one movement each could predict lower limb joint torques accurately. By incorporating transfer learning in inter-subject and inter-task scenarios, and thereby making use of knowledge learned from previous users/tasks, generalizability of both MLP and LSTM models improved. The implications of this is that since joint torque was accurately predicted with little training data from new subjects, this method should be feasible to predict torque even when new subjects, for example people with motor disorders, are unable perform a wide range of movements in the training process.

The developed virtual HMS makes it possible to compute torques, muscle impulse, contact forces and interactive forces involved in simulated movements with different knee exoskeleton assistive strategies and different weight distributions. The results of our simulation of RA, SPM and DPM control strategies

on exoskeletons suggest that both joint position and rotational direction should be considered in an assistive strategy, and that a damping component in a pendulum-based assistive strategy could be beneficial to further reduce the user's required muscle effort. We have also demonstrated that a higher assistive torque may incur a trade-off in the form of higher interaction forces, which should be kept in mind when designing a control strategy. Furthermore, no single physical design was optimal for all movements. A user's daily routine, activity goals and residual muscle strength should be considered while designing an exoskeleton's physical design and control algorithm. Detailed information at joint and muscle levels provided by using a musculoskeletal modeling environment can contribute to the design of exoskeletons.

A suggestion to further improve robustness of joint prediction models is to employ a Bayesian neural network that offers uncertainty quantification of the prediction that would be adopted to auto-detect such situations first and therefore guide the enrichment of training data to re-train a robust model. Another suggestion is to perform a subsequent sensitivity analysis to determine more specifically how many and which movements are optimal for re-training the model when transfer learning is employed, to increase feasibility without sacrificing prediction accuracy. Furthermore, different techniques for squat and other movements should also be considered, as they can represent a wider spread of muscle activation patterns. In the user-exoskeleton interface, the strap's tension, material and surface friction, as well as the soft tissue properties of the user's thigh and shank could be considered when modeling the interactive forces. Furthermore, while it is crucial, it still remains to be studied whether methods to predict torque from EMG data in patient populations with disorders involving motor control are reliable and valid, and even whether simulations with HMS can accurately represent deviating movement strategies. These, indeed, will be the real test of feasibility, validity and transferability of the methods in this thesis.

Acknowledgements

Here, I would like to express my sincere gratitude to all those who have supported me during the past four years, and especially to:

Prof. Lanie Gutierrez-Farewik, my main supervisor, offered me this great opportunity to start my PhD in a wonderful group and leaded me to an interesting and promising research field. You are so kind all the time and it is my luck to meet you as my supervisor. I still remember the moment when you wrote an email to notify us about the appropriate clothes for the ESMAC conference in Norway in 2017, which is so sweet. I also remember the super welcoming vibe the first time we met in front of our department. You are always supportive of our study, like summer/winter school, of our research, like staying up reading my manuscripts. Sincerely, I deeply appreciate all the patience, encouragement, guidance, inspiration and continuous support you have given me. Thank you for all!

Asst. Prof. Ruoli Wang, you are not only a co-supervisor, but also a real elder sister and a great friend. Thank you for all the patient guidance, inspiring discussions, sharing your knowledge, experiences and ideas in my research and life. It's my honor to know you here, thanks a lot!

Assoc. Prof. Christian Smith, my co-supervisor, for all the valuable discussions we had together, all the encouragements and help.

Yixing, my co-author, for the development of the knee exoskeleton CAD model that I could incorporate into a musculoskeletal model to do the search on the design of human-machine system. Also for putting me as the second author for the research on the weight distribution of a knee Exoskeleton influences on muscle activation during movements.

Yingbai and Prof. Zhijun Li, my co-authors, for the construction of preliminary artificial neural network model to predict join torque, as a part of results for the joint torque prediction performance comparison between physics-based NMS and modern machine learning models. It is a great time to discuss with you.

Acknowledgements

Asst. Prof. Xueyu Zhu, my co-author, for the great cooperation on design of the physics solver-informed artificial neural network model to predict joint torque. I learned a lot from you, and you always provided quick responses when encountering problems.

Davit, my co-author, for deep learning models constructed in the lower limb joint torque prediction, and the interesting discussions with you all the time.

Davíð, our previous lab engineer, for helping me with data collecting. I still remember the special time when we collected data from four subjects in one day or turned on three or four heaters in October. Thanks for all and it is a great time to work with you. All people who participated in the data collection. Without you, this thesis will not be possible.

All the past and present members at the department of engineering mechanics. Thank you for making our department a nice working atmosphere! My former and present colleagues, Chantal, Zhouyang, Yixing, Ben, Israel, Niklas, Xiaochen, Zhongzheng, Asta, Lara, Antea, and Fredrik for all the interesting discussions, Monday seminars, and afterwork activities we had together.

My dear friends, Linqin , Tianqi, and Ge for all the company, great joy and memories in daily life. Yazhou, shiyang, and Hui, for all the great time and trips; and Song, for the meals and enlighten; and Huisi, for the camping and climbing time. Huan and Jinrong, nice to be a roommate with you and the happy time we hung out together. Zhi, Kequan, and Lizhou for the wonderful meals and majiang time. Yingbai, Hang, and Wen for the notable guidance and talk about research and life. Yi, Lijuan, Xuelin, Ke, and Xuan, for all the support and care. (我亲爱的盆友们，姐，胖奇还有哥，在日常生活里度过的点点时光。吴彦祖和诗阳，一起去找灰教授玩耍啊，还有那些难忘的旅途时光; 松哥大佬啊，感谢平时对我们的照顾和点拨啊；还有慧思，那些愉快的露营和攀岩时光。欢和豆豆啊，我可爱的室友们，感谢让我有事没事蹭饭吃啊:)还有晃悠的时光。 大智，克权还有立洲，一起去老王家蹭饭和打麻将啊。英柏，苏航师兄和齐雯姐，感谢一起探讨的研究与生活。感谢二笨、老大、雪林儿、科长和萱萱有事没事听我唠嗑).

あかねちゃん、いつもサポトしてもらったこと、励ましてもらったこと、大感しています。我々のスウェデンと日本の旅が素晴らしい思い出となり、一生忘れません。

我亲爱的爸妈，谢谢您们一直以来对我无条件的爱与支持。感谢我的外公、大舅、舅妈、小姨、二哥二嫂等亲人们一直以来对我生活与学业的支持和鼓励。我知道不管发生什么，您们都一直在我身边。

I would like to acknowledge financial support from Promobilia Foundation (ref nr 18200, 18014, 18202, 19302, and 21302) and Swedish Research Council (2018-00750 and 2018-04902) and China Scholarship Council (CSC).

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